

NEXATECH

Project Alpha — System Architecture

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Overview

Project Alpha is NexaTech's internal Enterprise AI Assistant. It provides a unified interface for company documents, operational databases, collaboration data, and approved external research.

Application Stack

The frontend uses React and TypeScript. The backend uses Python and FastAPI. LangGraph provides stateful agent orchestration. PostgreSQL stores application and operational data. Redis provides caching and short-lived workflow state.

AI and Retrieval

The assistant uses an LLM provider abstraction so the underlying model can be changed without rewriting application services. ChromaDB stores document embeddings for semantic retrieval. Document metadata includes source, department, document ID, version, and access classification.

Agent Topology

The Supervisor Agent routes requests. The Planning Agent decomposes complex tasks. The RAG Agent searches internal documents. The SQL Agent executes approved read-only queries. The Web Search Agent performs approved external research. Reflection and Critic agents validate research quality before final reporting.

Request Flow

A user request enters FastAPI, is validated and authenticated, then passes to ChatService. ChatService invokes the LangGraph workflow. The graph updates shared state as agents and tools execute. The final response is returned through the API or streaming endpoint.

Reliability

Production services must expose health checks, structured logs, latency metrics, and error rates. Agent traces should be captured through the approved observability platform.

Security Boundaries

Agent tools must enforce authorization independently of the LLM. The SQL Agent receives a read-only database role. High-impact actions require explicit human approval.

Related Documents

See DOC-ENG-003 Deployment Guide, DOC-SEC-001 Information Security Policy, and DOC-SEC-003 Production Access Control.

This document is synthetic data created for development, RAG, agent orchestration, and evaluation testing.